WU, YU-HSUAN

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EDUCATION

National Yang Ming Chiao Tung University (NYCU) 2028 (expected)

Master of Arts in Linguistics

- Relevant Coursework (expected): Phonology, Computational Linguistics

National Taiwan University (NTU) 2026

Bachelor of Arts in Foreign Languages and Literatures

- Member of NTU Computational Linguistics Specialized Program
- Member of NTU Chinese-English Translation Program
- Relevant Coursework: NLP, Computational Linguistics, Phonology, Semantics

EXPERIENCE

Cathay Financial Holdings — DDT AI Taipei, Taiwan
Feb 2026 – Present

ML/AI Engineer Intern

- Optimized AI News agent orchestration logic, data ETL, LINE integration and AWS deploy architecture.
- Built a recording-ASR-based BRD auto-revising agent with human-in-the-loop audit.
- Deployed a speech-driven form-filling service using streaming LLM on AWS for enterprise usage.

National Taiwan University Taipei, Taiwan
Jun 2024 – Jun 2026

Research Assistant — Prof. Shu-hao Shih

- Designed perception experiment schema, prepared stimuli, and analyzed data for non-moraic schwa stress-bearing naturalness.
- Surveyed artificial-learning-experiment related phonology papers and prepared experiment stimuli using Praat and PsychoPy.

Taiwan AI Academy (AIA) New Taipei City, Taiwan
Jun 2025 – Aug 2025

AI Developer & Technical Support Summer Intern

- Architected a LINE LIFF App career counseling chatbot with an n8n-based RPA workflow and managed user data with Supabase.
- Supported Enterprise Cloud Platform education courses as technical support.

PROJECTS

LLM Reasoning Research — Self-Discover-NLI ROCLING Conference 2025

- Developed and presented the "Self-Discover-NLI" prompt engineering framework, adapting the three-step Self-Discover pipeline (SELECT → ADAPT → APPLY) specifically for Natural Language Inference (NLI) tasks.
- Conducted systematic experiments on the IMPPRES dataset (95 tasks per trigger type), evaluating GPT-4o's handling of clefts, conditionals, modals, and other linguistic triggers.
- Achieved ~10% overall accuracy improvement; performed in-depth error analysis identifying systematic model weaknesses.

BU Agent — Web-Based Two-Mode BRD Co-Drafting Assistant Cathay Financial Holdings, DDT AI

- Built an internal web app using LangGraph + PostgresSaver with a split layout (chat ↔ mode-aware workspace); BU Agent serves as the BU's single entry point (two modes) to draft v1.0 BRD, then hands off structured deliverables to the downstream BA Agent.

- Explore mode: 5-stage question bank (diverge → converge → challenge) with a live 5-dimension rubric workspace; emits a dual handoff (JSON + 200–400-word paragraph) that auto-transitions to Consult, compressing ideation to 5–10 min.
- Consult mode: Step 1 auto-drafts the BRD outline from per-BU YAML templates, tagging sections auto_filled / needs_round2 / placeholder; Step 2 interviews only needs_round2 with live-preview inline edit, keeping BU interview time to 20–30 min.
- Outputs v1.0 BRD + summary.json + full conversation_trace + flag_for_ba_review list; handoff notifies the BA Agent for review.

voice-notion — Slack 語音轉 Notion 機器人雲原生架構 AWS (ap-northeast-1 + us-east-1)

- Re-architected a single-instance Next.js service into a cloud-native ECS Fargate deployment with DynamoDB, SQS and EventBridge to enable horizontal scaling and zero-data-loss rolling updates
- Built a Slack bot that converts voice or text @mentions into confirmable Notion pages and tasks, plus a daily automated standup summary
- Ran web and worker containers from one image in a single Fargate task, differentiating roles only by start command (node server.js / node dist/worker.js)
- Split webhook handling so web verified signatures, enqueued to SQS and returned 200 within 3s while worker handled 30–120s transcription and LLM work
- Replaced 6 in-memory state stores with a DynamoDB single-table design (PK/SK, TTL) so state survived task restarts and was shared across containers
- Added atomic operations using UpdateItem with ConditionExpression (claimDraft) and PutItem with attribute_not_exists to prevent duplicate submissions across containers
- Configured SQS main queue plus DLQ with VisibilityTimeout=180 and maxReceiveCount=3 to persist, retry and isolate poison messages
- Built a speech-to-text pipeline using ffmpeg (16k mono 16-bit LPCM, 3200B/100ms chunks) into Amazon Transcribe streaming with multi-language zh-TW+en-US identification
- Created custom Transcribe vocabularies (voice-notion-zhtw / voice-notion-enus) synced from lib/vocabulary.ts and MEMBERS_JSON to improve recognition of member names and project terms
- Integrated Bedrock Claude Opus 4.8 via the AnthropicBedrockMantle client in us-east-1 using ECS task role SigV4 auth with no API key
- Migrated speech recognition from Bedrock Nova 2 Sonic to Amazon Transcribe after Sonic returned transcriptLength:0 across three attempts, as it is a speech-to-speech model
- Fixed a standup retry storm by changing the cron route to enqueue + immediate 200 and setting EventBridge target MaximumRetryAttempts=0, eliminating duplicate summaries
- Reworked deployment for compliance (Security Hub ECS.2/ECS.5) by moving tasks to dual-AZ private subnets with assignPublicIp=DISABLED, NAT egress, readonly root filesystem and ephemeral volumes
- Reused the existing aaron-dev ALB by adding a Target Group and a host-based rule (priority 20) on the 443 listener instead of provisioning new ALB/ACM/Route53
- Stored Notion / Slack tokens, CRON_SECRET and MEMBERS_JSON as SSM Parameter Store SecureString values injected into the task at startup
- Split IAM into execution and task roles, scoping the task role to DynamoDB/SQS, bedrock:InvokeModel, bedrock-mantle:* and transcribe:StartStreamTranscription
- Evaluated Lambda + API Gateway, self-managed EC2 and App Runner against ECS Fargate on persistent worker, cold start/ffmpeg, HA, ops cost and private-subnet criteria
- Documented cost estimate of ~$57/month excluding AI usage by reusing the existing ALB and NAT, with a documented lever to switch to claude-sonnet-4-6 via environment variable
- Wrote an operations runbook covering releases, rollback by task-def revision, log tailing, DLQ redrive, secret rotation and scaling
- Iterated the task definition from rev1 to rev9, each revision tied to a specific failure such as Mantle 404, unsupported output_config, Security Hub findings and duplicate standup triggers

SKILLS

Languages: Mandarin (Native), English (Fluent)

Programming: Python

Cloud & Infrastructure: AWS (Lambda, Bedrock, OpenSearch), Amazon Bedrock (Claude Opus 4.8 via AnthropicBedrockMantle client), Amazon DynamoDB (single-table design, on-demand, TTL), Amazon ECR (image storage tagged by git SHA / timestamp), Amazon EventBridge (classic Rule, cron + API destination with Bearer auth), Amazon SQS (main queue + DLQ, long-poll consumer)

Frameworks: LangGraph

Tools: Microsoft Office, AWS CLI (register-task-definition, create-service, update-service, logs tail), Docker (node:20-slim image with ffmpeg, –no-cache builds)

Certificates: TOEIC (Listening and Reading 985, Speaking 180, Writing 170)